

Dr. Rakesh Bhaskaran Nair

Advanced Manufacturing & Surface Engineering

Real-time material integrity for extreme environments.

Principal Investigator and Early-Career Research Leader | Dublin City University

Manufacture | Monitor | Predict | Extend Lifetime

29	1,260+	h-index 19	18	EUR 220k+
Journal articles	Citations	Google Scholar	First/corr. author	PI/Co-PI funding

RESEARCH PROGRAMME

01 - Advanced Manufacturing

Additive manufacturing, thermal spray, cold spray and advanced alloys for reliable components and functional surfaces.

02 - Intelligent Materials Diagnostics

Thermal/optical sensing, spectroscopy, process monitoring, materials informatics and powder qualification.

03 - Durability & Lifetime Extension

Coatings, wear, erosion, corrosion and extreme-environment degradation linked to component lifetime.

SELECTED PUBLICATIONS

- Spectral Emissivity Hysteresis in NiTi Powders. Optics & Laser Technology, 2026.
- Nickel-Dependent Emissivity Calibration of NiTi Powders. Measurement, 2025.
- CMAS Corrosion Attack on Thermally Sprayed TBCs. npj Materials Degradation, 2024.
- HEA Coatings Deposited by Thermal Spraying: A Review. Metals, 2023.
- Enhanced Wear Resistance of AlCoCrFeMo Coatings via Thermal Spray. Surface & Coatings Technology, 2024.

ACADEMIC POSITIONING

Independent PI developing a coherent programme at the intersection of advanced manufacturing, surface engineering, materials diagnostics and component durability/lifetime extension. The work connects manufacturing quality, real-time monitoring and durability science for engineering components.

Flagship PI Project

Principal Investigator - Research Ireland ARC Hub

Metal Powder Quality Intelligence Platform for Advanced Manufacturing | EUR 185,000 | 2026-2027

TEACHING & SUPERVISION FIT

- MSc teaching: IR Sensors & Thermal Diagnostics for Additive Manufacturing; Thermal Spray Manufacturing Technologies.
- Teachable areas: manufacturing processes; surface engineering; materials characterisation; AM diagnostics; tribology and failure analysis.
- Mentored 9+ postgraduate researchers and trained 10+ HQPs on laboratory safety and SOPs.

AVAILABLE FOR

Academic collaborations | Invited lectures | FDPs | MSc teaching | Curriculum development | External PhD/MSc co-supervision | Visiting/adjunct opportunities | Fellowship hosting

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